will comes i.e., [8] yr's.

Ex:-1:- 96th yr is leap year 100th yr is non-leap centuary year & 104th yr is leap year. So, 104-96 = 8 yr's gap b/w 2 leap yr's when in b/w then non-leap centuary year comes.

↳ Finally after knowing all the above synopsis points the conclusion is if we know no. of odd days upto given date of month of particular yr then we can find day on that date. Finding odd days is important.

Ex:-2:- What was the day on 01-01-01?.

Ans:- Up to 01-01-01 one day completed [when question given like this the day on given date is to be considered as completed to find it's day of week] means one odd day means monday.

01-01-01 → Monday.

Ex:-3:- What was day on. 01-02-01?

Ans:- Jan month & one day in feb completed means total 32 days completed : odd days $\frac{32}{7}$ = 4 odd days. means Thursday.

01-02-01 → Thursday.

Ex-4:- What was day on. 01-01-02?

Ans:- one non-leap year + one day completed. So,

(161)

So, odd days = $\frac{365+1}{7}$ => 52 odd days means "tuesday". (or)

Method 2:- Non-leap year next year 1st day is "+1" day.

01-01-01 → Monday then 01-01-02 ⇒ M+1 = Tuesday.

↳ When yr's given like the above solving easy generally.

But, when yr's with big completed years are given then,

Ex:4:- What will be the day on 26-08-2020?

Ans:- When the question given like this, the odd days are calculated from bigger unit to smalled unit of completed time & add all of them, find odd days & then day.

26-08-2020

- 2019 (Completed years)
  - 2000 Completed Centuries → 2000 is exactly 400 multiple then '0' odd days
  - 19 Completed Non-centuries
    - $\frac{19}{4}$ = 4 LY
    - 19 - 4 = 15 NLY
    - 4×2 + 15×1 = 23
    - $\frac{23}{7}$ => 2 odd days

In completed bigger unit of time is in completed centuries & non century yr's odd days are '0'+2 = 2 odd days

- 7 months Completed While finding odd days in months, Cross Check running year is leap or not.
  - if leap feb → 29
  - non-leap feb → 28
  - ↳ here is leap then. J + F + M + A + M

J	F	M	A	M
31	29	31	30	31.
3 odd	1	3	2	3

Ju + Jul

Ju	Jul
30	31
2	3

Total = 3+1+3+2+3+2+3 = 3 odd days

- Running month odd days: $\frac{26}{7}$ = 5, 5 odd days.

(162) Total odd days = $0+2+3+5 = \frac{10}{7}$ ⇒ 3 odd days.

⇒ day on 26-08-2020 is wednesday.

↳ Notes on odd days for centuries & Non-century years:-

i) If centuries are 400 multiple then odd days are zero.

ii) After observing completed centuries divide it with 400, if remainder '0' ⇒ odd days = 0

if remainder '100' ⇒ " " = 5

if " " '200' ⇒ " " = 3

if " " '300' ⇒ " " = 1

iii) if non century years observed in calculation, then odd days = Non-century years + No. of leap years in it.

Ex:- Odd days in 42 years:-

M(i) 42 years

$\frac{42}{4} = 10$ L.Y 42-10 = 32 N.L.Y

$10\times2 + 32\times1 = \frac{52}{7}$ ⇒ 3 odd days

(or)

M(ii) 42 years

odd days = $42 + \frac{42}{4}$ [Q]

= 42+10 = 52

⇒ $\frac{52}{7}$ ⇒ 3 odd days

Ex-2:- The day of week on 16-03-2024?

Step1:- Find out odd days in Completed years first

2023 = 2000 + 23 yrs = 0[400 multiple] + 23 + 5 ⇒ [0] odd

Step2:- Cross check running year is leap (or) Not. If leap consider 29 in feb otherwise 28 in feb.

Step3:- Find odd days in completed months.

J(31) → 3 + F(29) → 1 = 4 odd days.

Step4:- Find odd days in completed days.

M(16) days ⇒ $\frac{16}{7}$ ⇒ 2 odd days.

Total odd days = 0+4+2 = 6 ⇒ Saturday

Ex:-3:- What will be the day on 04-10-2098?

St:-1:- 2097 = 2000+97 = 0+97+24 = $\frac{121}{7}$ ⇒ 2 odd days

St:-2:- ~~J(3)~~ Running year non-leap year then feb → 28 days

St:-3:- J(3)+F(0)+M(3)+A(2)+M(3)+J(2)+J(3)+A(3)+S(2)

odd days in completed months = $\frac{21}{7}$ = 0 odd days.

St:-4:- Odd days in October = 4 odd days.

Total Odd days = 2+0+4 = 6 ⇒ 04-10-2098 is Saturday

Ex:-4:- Day on 23-04-2023?

Ans:- 2000 [0] + 22 [22+5] + Jan [3] + Feb [0] + Mar [3] + Apr [23]

⇒ 56 odd days ⇒ $\frac{56}{7}$ = 0 odd days ⇒

∴ 23-04-2023 is Sunday.

Ex:-5:- If today is Sunday what day will be after 71 days?

Ans:- Sun + $\frac{71}{7}$ [1] (odd day) = Monday

Ex:-6:- If today is Wednesday, then what day will be after 2900 years on same date & month?

Ans:- Wednesday + Odd days in 2900 years

= Wednesday + $\frac{2900}{400}$ ⇒ Rem = 100 ⇒ odd days = 5

= Wednesday + 5 days = Monday. (or)

= Odd day of wednesday = 3+5 days = $\frac{8}{7}$ ⇒ 1 odd ⇒ Monday